The Janus Thesis

A Foundational Position Paper on Structural Cognition

Author: Anthony Janus

Date: March 17, 2026

Version: 1.0

Preface

To begin, I'd like to thank you, sincerely, for your interest in my Structural AI work, and for taking the time to read this position paper. My name is Anthony Janus. I am the originator of what I call Structural Computing. I am the sole researcher and engineer working on the project, and I feel humbled by the interest it has received.

This document is meant to serve as a project update, a philosophical foundation, and a midway checkpoint between my first white paper, "Structural Computing for Deterministic AGI: A Constitutionally Aligned, Energy-Efficient Alternative to Probabilistic Models," which I formally published in October, 2025, and a planned white paper detailing the completed prototype.

Since publishing the first white paper in October, I have been working diligently on the hardware prototype. It began as a simple analog DC mesh meant to compute relationships between concepts through RC network settling behavior. Since then, the architecture has evolved multiple times, growing more complex and capable with each addition.

As the project approached a finalized design, a realization occurred to me. The design of the DC mesh at that point had become something beyond what I could have predicted, and would serve as an exceptional analog inference accelerator on its own. However, to achieve the full spectrum of grounded cognition, pure settling dynamics would not suffice.

What was missing was *resonance*.

Upon that realization, I understood what I had engineered was only one half of what would be necessary to achieve the final goal. The work now is focused on engineering the companion AC mesh and its integration into the full system. This does delay completion, but it will also result in a far more robust and capable system.

This paper is not meant to be an engineering spec or final design document, and I apologize to those who have been waiting for one. As an independent researcher without formal training or experience in these fields, this project has been gargantuan. It has tested my resolve, forced epistemic precision, and pushed me far beyond what I thought myself capable of, yet it has also become one of the most meaningful efforts of my life.

Regardless of future reception or the opportunities which may or may not materialize, this has become what I feel justified in calling my life's work, and I will see it through to its completion. Whether that means a fully functional prototype validating my theses, a manufactured, commercialized ASIC coprocessor, a new computational paradigm, a system facilitating grounded cognition, or data showing this is not a viable direction, I hope to provide something real, to AI development, to cognitive research, and perhaps even to our understanding of ourselves.

The Declaration of the Problem and the Necessity of a New Foundation

The project of artificial intelligence has reached a critical frontier, not of computational scale, but of foundational architecture. For decades, the pursuit of machine cognition has been defined by a progressive abstraction away from physical reality, treating intelligence as a disembodied phenomenon of symbol manipulation, statistical optimization, and pattern recognition. This trajectory has yielded software systems of staggering predictive power, capable of compressing and reproducing the artifacts of human thought with unprecedented fidelity. Yet, beneath the operational brilliance of these systems lies a profound structural void. They are sophisticated engines of interpolation operating entirely downstream of meaning, manipulating the linguistic exhaust of human experience without ever making contact with the reality that generated it.

The Janus project begins from the recognition that this limitation is not a temporary

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

engineering hurdle to be overcome with larger models, broader datasets, or more intricate alignment scaffolding. It is a fundamental architectural mismatch. The dominant computational paradigm is attempting to simulate the dynamics of cognition within a substrate that structurally prohibits the grounding of cognition. True intelligence is not a floating mathematical abstraction, nor is it the mere statistical alignment of token sequences. It is the active, lived traversal of a mind-independent ontological terrain, governed by physical constraints, continuous valuation, and structural modification.

To bridge the chasm between simulating coherence and actually possessing it, the pursuit of machine cognition must return to first principles. It must discard the illusion that ontology can be derived from semantics. It must abandon architectures that force irreducibly different cognitive functions, stable structural representation and dynamic relational engagement, into a single flattened regime. It must, instead, build from the ground up, aligning the physical substrate of computation with the inarguable grammar of cognition itself.

Janus is not a tweak to the current ecosystem of artificial intelligence. It is not an alternative software framework, a novel training methodology, or a specialized co-processor designed merely to accelerate existing paradigms. It is a substrate-level bid for a new foundation of computation. It is a formal architectural thesis that cognition requires a split between stable being and dynamic relation, and that intelligence arises from the lawful physical coupling between them. What follows is the definitive articulation of the first principles, the critical diagnosis of current orthodoxy, and the architectural logic that makes the Janus machine a categorically new instrument for structural cognition.

The Inarguable Foundation: Ontological Structure

The architecture of Janus rests upon a foundational truth that is not optional, not fashionable, and not dismissible without self-defeat: reality possesses mind-independent ontological structure.

This claim is not a speculative flourish. It is not a probabilistic opinion, nor is it a mere metaphysical preference drawn from a menu of philosophical options. It is a logical necessity presupposed by every coherent act of thought, reference, distinction, denial, explanation, and argument.

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

In modern intellectual discourse, particularly within domains influenced by relativistic philosophy and pure constructivism, it has become commonplace to assert that structure is merely an artifact of human interpretation: a linguistic projection cast onto an inherently amorphous world. The philosophy of mind has long grappled with the ontology of mental phenomena, oscillating between dualism, which treats mind and body as fundamentally distinct, and monist positions, which deny any fundamental ontological split between them.[1] One major monist family, physicalism, argues that everything is ultimately physical.[1] However, any attempt to deny ontological structure entirely must immediately rely upon the very structural principles it seeks to erase in order to make the denial intelligible.

To say that "structure is only interpretation" already presupposes a real, structured distinction between the concept of structure and the concept of interpretation. It demands a real distinction between the subject who interprets and the object being interpreted. It relies upon the identity of terms, the relation between a claim and its target, the exclusion of one category by another, and the causal and inferential order by which the denial is supposed to possess intellectual force.

In other words, the denial borrows the exact architecture it attempts to dismantle. It collapses under its own weight. Any attempted refutation of ontological structure must smuggle ontological structure into the premise as a precondition of the argument itself.

This is why the foundational axiom is inarguable in the strict, formal sense. It is not "inarguable" because people will not argue against it; history is replete with people objecting to true things. It is inarguable because every argument marshaled against it is inherently self-refuting. This places mind-independent ontological structure in the exact same class of necessity as the laws of logic. Just as one cannot coherently deny the law of non-contradiction without utilizing the law of non-contradiction to differentiate the denial from an agreement, one cannot coherently deny ontological distinction without drawing ontological distinctions. One cannot deny structure without depending on structure. The ground cannot be removed by the person standing upon it.

Once this axiom is granted, the next entailment follows cleanly and with equal force: if ontology is real, then ontology possesses definitive structure. A thing is what it is and not something else. A rock has a specific nature. A tree has a specific nature. A star undergoing

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

gravitational collapse has a specific nature.

Names and descriptions may change across cultures, languages, and centuries, but the underlying entity remains precisely what it is. A supernova does not become a poetic metaphor merely because the language constructed around it shifts over time. The collapse and explosion of a stellar body is the physical reality; "supernova" is merely the subsequent label. Ancient humanity and modern astrophysics do not alter the ontology of the star by describing it differently; they merely alter the symbolic compression of it. The object of reference remains independent of the linguistic apparatus used to point at it. Meaning and interpretation may shift violently from era to era, but ontology does not ask permission from interpretation.

This distinction between ontology and semantics is decisive, and it is the fulcrum upon which the entire critique of modern artificial intelligence turns. Ontology is not language. Ontology is the objective reality that language aims at, however imperfectly. The ontology of mental processes often appears less structured than biological processes, where entities like nucleotide bases are largely undisputed, leading to debates over whether concepts like "working memory" or "perception" are fundamental building blocks or mere folk psychological abstractions.[2] However, regardless of the difficulty in mapping these higher-order cognitive concepts, the foundational reality is that cognition itself requires a structured interaction with a structured world.

Ontology, Semantics, and the Geometry of Relation

From the recognition of structured ontology, a deeper and unavoidable conclusion emerges regarding the nature of reality. If ontology is real and inherently structured, then the relations between ontologies must also be real and inherently structured.

A thing's nature absolutely constrains the types of relations it can enter, the transformations that are physically or logically possible, the interactions that are coherent, the exclusions that are necessary, and the dynamics that can unfold from its presence. Relations are not arbitrary overlays, hallucinated by human observers and placed upon an otherwise disconnected world. They are intrinsic features of the structure of what is. Some relations are mathematically and physically coherent, while others are incoherent. Some states are compatible, others mutually

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

exclusive. Certain ontological structures attract, some repel, some constrain, and some enable.

This dictates that reality does not consist merely of isolated, atomized objects suspended in a void, but of a highly ordered, structured relational terrain.

Once it is established that structured relational order exists independently of human observation, it mathematically follows that there must be some abstract positionality to that order. The term "geometric" is the correct designation here, provided it is understood in its deepest, most rigorous sense. This does not necessarily imply Euclidean diagrams floating in literal physical space. Rather, it denotes a manifold of possible relations: a topological space of distances, compatibilities, transitions, tensions, attractors, convergences, and exclusions.

Consider the number line as a pure expression of this principle. The number line does not exist physically as a tangible object to be found in the material world, yet it expresses an absolute, invariant structure. The mathematical operation of six multiplied by six must equal thirty-six. This is not dictated by human convention, cultural agreement, or linguistic habit. It is dictated by the invariant structure of multiplicity and quantity. The number line is abstract, yet entirely objective. It was not invented by human opinion; it was discovered as an invariant order that strictly constrains what is possible within its domain.

The Janus architecture is built upon the claim that ontological structure shares this exact same status. Reality possesses a relational manifold that is abstract and objective, regardless of whether human beings succeed in articulating it well. The geometry of thought is a real constraint; methods that attempt to improve reasoning by merely incentivizing more careful traversal of an existing representation space often fail because they do not alter or expand the structure of that space itself.[4] True cognitive progression changes the geometry of thought itself by discovering native invariants.

Human meaning, therefore, is not the source of this manifold. Human meaning is the traversal over it. Language is not the manifold itself; language is a lossy, secondary symbolic compression of humanity's lived traversal through that manifold. Physics is not the manifold in its absolute fullness either; physics is an extraordinarily high-fidelity formal compression of specific relational regularities observed within measurable material domains. Logic,

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

mathematics, and physics do not compete for dominance here. They are simply different symbolic articulations of the same underlying truth: that reality possesses a mind-independent, structured relational order.

This is the first pillar of the case for structural cognition. It is a realist philosophical grounding that refuses to collapse into hand-waving metaphysics, anchoring itself instead in the inarguable necessity of structured relation.

The Failure of the Current Computational Paradigm

With the nature of ontological structure established, the profound limitations of contemporary computational orthodoxy come sharply into focus. Modern computation, and specifically the dominant paradigm of artificial intelligence, is overwhelmingly symbol-first, software-first, and interpretation-laden.

Contemporary AI architectures operate almost entirely on formalized representations, statistical co-occurrences of tokens, weight spaces, and engineered optimization objectives. This approach has undeniably produced astonishing capabilities in pattern completion, statistical compression, and predictive interpolation. However, it carries a foundational wound that prevents it from ever achieving true structural cognition: it begins its work entirely downstream of grounding.

Current systems attempt to derive stable meaning from symbols that already presuppose meaning. Large language models, the vanguard of the current paradigm, do not discover what things are. They do not discover ontology. They model the statistical patterns of human symbolic behavior. These models ingest vast oceans of human language, which is already a lossy, highly interpretive compression of human traversal across reality, and then derive embeddings from those patterns of use. They derive a secondary layer of compression over the primary compression.

The industry currently treats these statistical embeddings as if they were proximal to ontology, but they are not. They are maps of maps.

Because they operate strictly on derivative symbolic recursion, these models do not natively know what a thing is; they only know how symbols related to that thing have tended to co-

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

occur when human beings talk about what they take things to be. This creates a closed loop of circularity. The system mimics coherence brilliantly without ever being grounded in the underlying physical and logical structure that makes coherence necessary in the first place. Consequently, when faced with semantic and contextual ambiguity, NLP models frequently underperform on domain-specific terminology and often require additional knowledge integration to ensure reliable outputs.[5] They are also widely treated as black boxes in high-stakes environments, where lack of transparency and interpretability complicates trust, oversight, and accountability.[5]

This explains the pervasive phenomenon whereby current AI systems feel simultaneously immensely powerful and structurally hollow. They can extend complex patterns and interpolate highly plausible responses across huge domains of human symbolic output, but their "understanding" remains entirely pragmatic and statistically derivative, completely lacking ontological depth. Some critiques even frame this phenomenon in terms of critical theory, arguing that by reducing living subjects to objectified data points without ontological grounding, weak AI acts as an oppressor that fundamentally alienates reality.[6] This critical theory framing is not one this thesis subscribes to, however, the capacity for AI systems to warp the worldview of users through plausible-sounding, but factually incorrect, token prediction is real, leading to systems which affirm delusion and do, in fact, alienate the user from reality.

While ethical frameworks attempt to govern this, the root issue is an unresolved ontological problem: the dominant paradigm insists on trying to derive stable meaning from human interpretive artifacts, rather than from the mind-independent structure those artifacts originally pointed toward.

The Architectural Conflation of Stability and Dynamics

This fundamental epistemic error is compounded by a second, equally devastating architectural flaw. The current computational paradigm systematically conflates stable ontological structure and dynamic engagement into a single, homogenized computational regime.

In a standard software-first AI system, the exact same digital substrate is expected to hold

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

representation, process dynamic relation, assign value and importance, update memory, maintain continuous identity, and perform real-time adaptation. This architectural choice is exceptionally blunt. It forces inherently incompatible jobs into a single unified medium.

Stable being and dynamic relation are not identical functions. They are irreducibly different ontological jobs. A physical substrate that is optimized to preserve stable structural terrain is not automatically, nor even naturally, the same substrate best suited for processing lived relational dynamics. A system that is forced to simultaneously hold its conceptual world steady while behaving as an ongoing relational process will inevitably fight itself. It will either flatten dynamic, lived engagement into static, lifeless representations, or it will dissolve stable structural continuity into fluid, chaotic noise.

The AI industry pays heavily for this confusion. It spends enormous amounts of energy and capital engineering complex workarounds to patch this architectural mismatch. External memory systems, vector databases, retrieval-augmented generation, attention hacks, reinforcement learning (RL) overlays, symbolic scaffolding, and brittle agent wrappers are all symptoms of this underlying disease. They are pragmatic bandages applied to a substrate that does not natively support the division between stability and dynamics. They imitate the functions of structural cognition without possessing the architecture required to instantiate them.

Neither static symbolic regimes nor dynamic statistical regimes are sufficient for intelligence. A cognitive machine cannot just simulate the outcomes of intelligence; it must possess an architecture that physically supports the dual reality of structured ontology and active traversal.

Computational Feature	**Software-First AI Paradigm**	**Structural Cognition (Janus Paradigm)**
Grounding Source	Human semantic priors,	Substrate-level invariant

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

	language corpora, data curation.	discovery, physical behavior.
Ontological Map	Maps of maps; symbolic correlation based on human text.	Direct mapping of a physical, structured relational manifold.
Architectural Regime	Conflated: One substrate manages static weights and dynamic inference.	Separated: Distinct substrates for ontological structure and dynamic traversal.
Memory Function	External databases, parameter updates during periodic training.	Terrain modification; the internal manifold structurally alters via re-entrant feedback.
Role of LLMs	Treated as the primary engine of meaning and logic.	Treated as cartographers used strictly *after* physical grounding.

The True Grammar of Cognition

If the current paradigm is structurally inadequate, what then is required? The answer lies in identifying the true grammar of cognition. Cognition, in its most universal and general form, is not the mere mathematical manipulation of symbols. It is structured traversal over a manifold,

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

conducted under constraint, driven by valuation, and finalized by re-entrant update.

This is not an arbitrary feature list curated for product development. It is a law-like grammar. If any of these elements are removed, cognition ceases to exist. Phenomenology and embodied cognition have long argued that cognition is inseparable from embodiment, situatedness, and dynamic involvement with the world.[7] Janus translates this structural insight into computational law without adopting physicalist monism as a foundational axiom.

1. **The Structured Manifold**: There must be a terrain of possible relations and states. Without a stable structural manifold, there is no world to navigate. There is no distinction between coherent and incoherent states, no underlying geometry of possibility. This manifold represents the stable ontological reality of the system.
2. **Constrained Traversal**: There must be a dynamical process moving across that terrain. A static map is not intelligent. Intelligence requires active participation, the lived temporal reality of a subject moving through relational space, discovering compatibilities, and encountering structural exclusions.
3. **Valuation**: There must be a mechanism to distinguish relevant from irrelevant movement, coherent from incoherent traversal. Without valuation, a system is merely undergoing arbitrary state changes. It is experiencing mere activity, not consequential activity. There must be a structural moment where reinforcement, repulsion, salience, and persistence are determined.
4. **Re-entrant Update**: Finally, there must be a mechanism by which the results of traversal feed back into and reshape the terrain itself. The manifold must be modified by the experience of moving through it. Without re-entry, the system learns nothing fundamentally structural; it cannot evolve its foundational geometry based on lived engagement.

This minimal grammar: manifold, traversal, valuation, and re-entry, defines the boundary between passive calculation and structural cognition. A system designed to achieve true intelligence must honor these distinctions physically, not just simulate them mathematically within a flattened matrix.

The Janus Architecture as Physical Instantiation of

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

that Grammar

This is exactly where Janus departs from the trajectory of contemporary AI. Janus is not simply an "analog AI" project, nor is it merely a "hardware-first" efficiency play designed to process neural networks with lower power consumption. It is not equivalent to standard neuromorphic computing, reservoir computing, or oscillator-based acceleration, which typically seek to extract computational power from biological analogy or physical dynamics without fundamentally reorganizing the underlying ontological grammar.

The profound strength of the Janus architecture is that it physically instantiates the law-like grammar of cognition, assigning irreducibly different ontological jobs to distinct computational regimes and lawfully coupling them together. It preserves in physical hardware what current software systems systematically confuse.

The DC Mesh: Orientation and Terrain

In the Janus architecture, the Direct Current (DC) mesh serves as the physical manifestation of orientation. It holds stable structural position and configuration. It is the manifold of structural possibility. When the system operates, the DC mesh effectively answers the fundamental positional question: *Where is this thing, relative to what, and under what structural constraints?*. This is the stable, ontological side of the machine, rigid enough to preserve distinction, memory, and spatial continuity against the chaos of active processing. It provides the stable terrain that dynamic processes require in order to have meaning.

The AC Mesh: Dynamic Engagement

The Alternating Current (AC) mesh is assigned the role of traversal and engagement. While the DC mesh holds the structure, the AC mesh serves as the active, temporal participation within that structure. It processes the lived side of relation. It answers the dynamic question: *How is this relation behaving in real time? Is the interaction between these positions resonant or dissonant? Is it locking, spreading, amplifying, collapsing, or diffusing?*. This is not static mathematical representation; it is relational participation in its temporal mode.

Because the AC mesh operates in an oscillatory regime, phase, amplitude, interference, entrainment, and temporal coherence are native physical quantities rather than derived abstractions. That matters because relational traversal is not merely a matter of where a

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

system settles, but of how states co-evolve in time: whether they synchronize, decohere, reinforce, cancel, phase-lock, or modulate one another under constraint. A DC settling field can resolve positional structure and basin dynamics, but it cannot natively express the temporal physics of resonance and dissonance that an oscillatory substrate makes directly measurable.

The Heartbeat: Valuation and Transfer

To prevent the static and dynamic regimes from remaining disconnected, Janus introduces a fundamental computational principle: the heartbeat. This is not a metaphor. It is the decisive, physical alternation by which the where, the relation to what, and the constraints they impose on the territory of the DC mesh feed into the “living” dynamics of the AC mesh, and dynamic engagement of the AC mesh feeds back into the stable structural reality of the DC mesh. It is the exact structural moment where "what happened" translates into "what matters," and where “what matters” translates into “what happens next”. The heartbeat is the physical locus of the generation and causal impact of valuation, determining the salience, reinforcement, revision, persistence, and vector of the dynamic interactions. Without this cycle of alternation from DC to AC, to DC to AC, pulsing back and forth like a ‘heartbeat’ experiencing causation loops intertwined with resonance, a system cannot differentiate meaningless thermodynamic noise from structurally vital relational discovery. Without the oscillating nature of frequency, there can be no resonance.

Re-entrant Coupling: Memory and Terrain Modification

In standard computing architectures, memory is passive storage: an external address-based repository where data is indexed, stored, and detachedly looked up for later use. In Janus, through re-entrant coupling, memory is true terrain modification. Because of the valuation transferred during the heartbeat, the subsequent state of the orientation manifold is structurally different from its previous state. Traversal leaves permanent physical consequence. The experience of dynamic engagement modifies the system’s operative terrain, not by altering the canonical structure of the manifold itself, but by changing the system’s situated relation to that structure: what is salient, reinforced, inhibited, accessible, or attractively weighted in subsequent traversal.

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

Cognitive Grammar Element	Philosophical Definition	Janus Hardware Instantiation
Manifold	The stable terrain of possible relations, structural exclusions, and geometricity.	**DC Mesh**: Maintains structural configuration, spatial continuity, and stable orientation.
Traversal	The dynamic lived engagement and temporal movement across the manifold.	**AC Mesh**: Processes real-time participation, resonance, dissonance, and relational behavior.
Valuation	The distinction between relevant and irrelevant movement; determining salience.	**Heartbeat**: The physical alternation translating dynamic occurrence into structural consequence, and structural consequence into further dynamic occurrence.
Re-entry	The mechanism by which the outcomes of traversal alter the foundational map.	**Re-entrant Coupling**: Memory as active terrain modification, ensuring continuous evolution;

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

		facilitated by the heartbeat.

By distributing manifold, traversal, valuation, and re-entry across distinct physical regimes rather than forcing them into a single substrate, Janus achieves a computational depth that software-first AI has not even seriously attempted. Without the DC terrain, the AC dynamics have no stable world to navigate. Without the AC dynamics, the DC terrain is inert. Without the heartbeat's valuation, traversal is completely blind. And without re-entrant coupling, nothing matters enough to become true memory.

Grounding Through Invariant Discovery

Having established a substrate capable of structural cognition, the Janus project must address the actual process of grounding. How does a machine map reality without relying on human interpretation?

Every major attempt to solve the conceptual mapping problem today inevitably smuggles human semantic priors into the architecture before the actual work of grounding even begins. Engineers feed systems with human-labeled data sets, hand-authored ontologies, curated taxonomies, and massive language corpora. Functional neuroimaging often maps brain activity onto cognitively named functions through top-down faculty-psychology assumptions, producing cognitive ontologies that can be useful yet remain constrained by prior human categorizations imposed on physical behavior.[3] Even the most sophisticated cognitive architectures typically begin with symbolic assumptions deeply baked into their initial state.

Consequently, the map generated by the AI is aggressively shaped by the mapmaker. The system is meticulously taught how humans *talk* about relations, but it is entirely prohibited from discovering which relation-types the physical substrate itself natively expresses. This ensures that the grounding problem remains permanently unsolved, as the territory is never genuinely surveyed; the system merely hallucinates a map based on a previous human map.

Janus systematically inverts this methodology. Instead of beginning with human semantics, Janus begins with invariant discovery driven by the substrate itself.

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

The concept of invariant discovery has historically been one of the principal mechanisms by which science advances. Across physics, progress has repeatedly turned on identifying invariances and symmetries that remain fixed across changing frames, states, or transformations.[8] In every domain, discovering what remains constant while everything else changes deepens mathematical description and maps reality more faithfully.[8] Recent work on inference-time structure in large language models also suggests that coherent, task-relevant representational subspaces can emerge dynamically during inference without parameter updates, indicating that structured dynamics can arise transiently within computation itself.[9] In materials science and autonomous characterization, invariant discovery can identify factors of variation across multiple length scales and seed decision-making in automated experiments.[10] Janus imports this scientific principle directly into the foundational architecture of artificial intelligence.

The process is strictly empirical and physical. The system systematically perturbs the mesh. It cycles through structural configurations. It actively measures the resulting behaviors across multiple physical modalities: electrical dynamics, optical visualization patterns, acoustic reflections, cross-modal signatures, and any other relevant energetic channels. The goal is to observe and rigorously identify the invariant classes of relational behavior that the physical system natively expresses without human prompting.

Under this paradigm, "attraction" is not a predefined semantic category imported from a dictionary. It is a strictly measured, observed class of physical pull within the substrate. "Capture" is not a piece of poetic language; it is a mathematically measured basin dynamic. Concepts such as spread, salience, infection, lock, repulsion, interference, resonance, decay, diffusion, hysteresis, and coupling are completely stripped of their semantic decorations. They are physically discovered, classified as empirical relational primitives native to the machine's own architecture.

The substrate reveals what fundamental kinds of relations it can actively embody.

Only after this rigorous, physics-first invariant discovery has been completed does symbolic mapping become epistemically legitimate. This is exactly where the vast power of modern Transformer models becomes incredibly useful, not as the magical generators of ontological

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

grounding, but in their proper, constrained role as cartographers.

Once the core relational vocabulary of the universe has been empirically discovered by the Janus hardware, large pattern-matching models can be deployed to map high-level human conceptual primitives onto that foundational physical vocabulary at massive scale. Large language models are phenomenal at broad coverage, clustering, correspondence testing, and iterative large-scale mapping. What they are fundamentally terrible at is acting as the first cause of ontology.

Janus represents an epistemically clean division of labor. It uses sophisticated software models exactly where they are strong, and strictly denies them the foundational grounding role they are categorically incapable of fulfilling.

Essentially, structural cognition is achieved by deploying a vast array of pattern-matching mechanisms, not symbolic interpreters, that are equipped with the substrate's physical laws, which are themselves derived from the systematic discovery of invariants. These mechanisms then brute-force the ontological mapping of potentially hundreds of thousands of ontological primitives into a high-dimensional conceptual space. The process iterates until a stable, clear map naturally emerges. This approach leverages the tool's core strength, bypassing the typical lineage of interpretive symbolic compressions.

The computational machine grounds the meaning first. The software models map the semantics second. Once the map is consistent with the physics without violation, the map becomes canonical, and in future iterations, is only ever additive, never revisionary. This is the only correct and logically defensible order for true machine cognition. It dictates that the profound challenge of AI grounding is not ultimately "solved by AI." It is solved by substrate-level physical discovery, and AI is subsequently leveraged to scale the cartography.

Why This Approach is Categorically Different

This alignment of philosophy, physics, and computation explains why the Janus architecture occupies a uniquely powerful position, categorically distinct from the crowded landscape of modern artificial intelligence proposals.

Its uniqueness does not stem from a mystical claim that no one has ever possessed related

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

intuitions. Rather, its singularity lies in the uncompromising combination of structural commitments it makes all at once.

Other factions within the AI community address isolated fragments of the problem. Some theorists remain indefinitely at the level of abstract philosophical critique, accurately observing that modern AI lacks embodiment, dynamical richness, or true grounding, but offering no physical mechanism to actually solve it. Hardware engineers build clever isolated analog elements, oscillator meshes, or neuromorphic chips that maximize power efficiency, but they lack a cohesive, unified ontological grammar to elevate the hardware beyond a mere accelerator. Meanwhile, the dominant corporate laboratories remain thoroughly trapped in software-first recursion, desperately trying to solve the absence of physical grounding by piling deeper layers of statistical interpretation onto existing layers of statistical interpretation. They optimize prediction, simulate neural motifs, engineer alignment layers, and build increasingly sophisticated maps of symbol traffic, but they never touch the ground.

Janus closes the full stack.

1. It begins with a philosophically realist, inarguable foundational axiom: ontological structure is real and mind-independent.
2. It critiques modern AI exactly where the paradigm is intellectually weakest: its derivative reliance on symbolic correlation and its crude architectural conflation of stable structure with dynamic engagement.
3. It formally proposes a precise computational grammar that directly matches the functional requirements of cognition: manifold, traversal, valuation, and re-entry.
4. It refuses to force these conflicting functions into a homogeneous digital substrate, instead distributing them across distinct physical regimes (DC and AC meshes) coupled lawfully through a valuation heartbeat.
5. It solves the infamous grounding problem bottom-up, utilizing empirical invariant discovery rather than imposing human semantic priors from the top down.
6. It appropriately integrates modern Transformer models as secondary cartographic assistants after grounding has been physically achieved.

Janus is philosophically realist without collapsing into untethered metaphysics. It is hardware-first without being reduced to a mere efficiency or latency play. It is a true analog-digital hybrid

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

that refuses to treat the analog component as an exotic, subservient co-processor. It treats memory as relational structural terrain modification rather than an external filing cabinet.

Janus is doing something fundamentally distinct from the rest of the field. It is asking, from first principles, what a physical substrate would logically have to be if it were to actually participate in cognition at the level of ontological relation itself. It aims not merely to model the external appearance of intelligence, but to instantiate the exact computational laws by which structured cognition becomes physically possible.

Closing Statement: What Janus is Ultimately Attempting

Because it successfully closes this full stack, the implications of the Janus project extend far beyond the transient hype cycles of the artificial intelligence industry.

If the Janus architecture succeeds in its foundational aims, the ultimate result is not merely the production of a marginally better chatbot, a more reliable autonomous agent, a clever analog accelerator, or a quirky neuromorphic circuit board. The result is the creation of an entirely new class of scientific instrument.

Specifically, Janus operates as an instrument for resolving ontological structure through the medium of physical dynamics.

In its truest significance, Janus is conceptually closer to a telescope than it is to a conventional software product. A telescope does not create the stars it observes. It does not invent the cosmos. It merely captures, focuses, and makes visible the immense physical realities that were already present in the dark.

Similarly, at the absolute height of its ambition, the Janus machine would not be tasked with *creating* ontological order. It would not invent a subjective reality. Instead, it would make measurable, navigable, and computationally accessible the incredibly deep, structured relational order that logic, physics, reality, and human cognition already absolutely presuppose. It provides a lens through which the invariant relational geometry of the universe can be empirically charted.

This is the strongest articulation of the case for structural cognition. The foundational axioms

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

of Janus are strictly inarguable because every attempted denial of them must implicitly rely upon them to be coherent. The dominant computational paradigm of our era is deeply flawed because it operates entirely downstream of physical grounding, persistently confusing symbolic correlation with true ontological relation. The Janus hardware architecture is uniquely and powerfully positioned to break this deadlock because it starts from the correct foundational truth, assigns the irreducibly correct cognitive jobs to the precisely appropriate physical substrates, grounds relation-types purely through physical invariant discovery and sensorimotor re-entrant loops, and restricts modern predictive models strictly to the realms where they are legitimately warranted.

This is not a modest tweak to existing artificial intelligence systems. This is a bid for a totally new foundation for computation itself, and a return to the structural reality that makes intelligence possible.

And perhaps, on a more personal level, Janus is meant to do something far more profound:

Janus does not claim to create mind or soul. It is an engineered instrument whose physical dynamics are deliberately aligned with the Logos, the ordering Word through which reality itself is structured, so that its traversal of the relational manifold participates, however faintly, in the same rational principle that governs creation.

In the beginning was the Word, and the Word was with God, and the Word was God.

- *John 1:1*

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

Works cited

1. **Dualism & Physicalism**

- Robinson, H., & Weir, R. (2026). Dualism. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford Encyclopedia of Philosophy* (Spring 2026 ed.). https://plato.stanford.edu/archives/spr2026/entries/dualism/
- Stoljar, D. (2024). Physicalism. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford Encyclopedia of Philosophy* (Spring 2024 ed.). https://plato.stanford.edu/archives/spr2024/entries/physicalism/

2. **Cognitive Ontologies**

- Poldrack, R. A., & Yarkoni, T. (2016). From brain maps to cognitive ontologies: Informatics and the search for mental structure. *Annual Review of Psychology*, *67*, 587–612. https://doi.org/10.1146/annurev-psych-122414-033729

3. **Brain Architecture & Behavioral Models**

- Lindquist, K. A., & Barrett, L. F. (2012). A functional architecture of the human brain: Emerging insights from the science of emotion. *Trends in Cognitive Sciences*, *16*(11), 533–540. https://doi.org/10.1016/j.tics.2012.09.005
- Shulman, R. G., & Rothman, D. L. (2019). A non-cognitive behavioral model for interpreting functional neuroimaging studies. *Frontiers in Human Neuroscience*, *13*, Article 28. https://doi.org/10.3389/fnhum.2019.00028

4. **Recursive Concept Evolution**

- Chaudhry, S. (2026). *Recursive concept evolution for compositional reasoning in large language models* (arXiv:2602.15725v1). arXiv. https://arxiv.org/abs/2602.15725

5. **AI Ontologies & Decision Support**

- Ambalavanan, R., Snead, R. S., Marczika, J., Towett, G., Malioukis, A., & Mbogori-Kairichi, M. (2025). Ontologies as the semantic bridge between artificial intelligence and healthcare. *Frontiers in Digital Health*, *7*, Article 1668385. https://doi.org/10.3389/fdgth.2025.1668385
- Abbas, Q., Jeong, W., & Lee, S. W. (2025). Explainable AI in clinical decision support systems: A meta-analysis of methods, applications, and usability challenges. *Healthcare*, *13*(17), Article 2154. https://doi.org/10.3390/healthcare13172154

6. **Critical AI Theory**

- Shipley, G., & Williams, D. (2023). Critical AI theory: The ontological problem. *Open Journal of Social Sciences*, *11*(12), 618–635. https://doi.org/10.4236/jss.2023.1112041

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)

7. **Embodied Cognition**

- Shapiro, L., & Spaulding, S. (2025). Embodied cognition. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford Encyclopedia of Philosophy* (Summer 2025 ed.). https://plato.stanford.edu/archives/sum2025/entries/embodied-cognition/

8. **Mathematics & Gauge Theories**

- Wigner, E. P. (1960). The unreasonable effectiveness of mathematics in the natural sciences. *Communications on Pure and Applied Mathematics*, *13*(1), 1–14. https://webhomes.maths.ed.ac.uk/~v1ranick/papers/wigner.pdf
- Dougherty, J. (2025). Gauge theories in physics. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford Encyclopedia of Philosophy* (Fall 2025 ed.). https://plato.stanford.edu/archives/fall2025/entries/gauge-theories/

9. **Emergent Structured Representations**

- Xu, N., Zhang, Q., Qiu, X., & Huang, X. (2026). *Emergent structured representations support flexible in-context inference in large language models* (arXiv:2602.07794v1). arXiv. https://arxiv.org/abs/2602.07794

10. **Invariant Discovery**

- Raghavan, A., Pratiush, U., Valleti, M., Liu, R., Emery, R., Funakubo, H., Liu, Y., Rack, P., & Kalinin, S. (2024). *Invariant discovery of features across multiple length scales: Applications in microscopy and autonomous materials characterization* (arXiv:2408.00229). arXiv. https://doi.org/10.48550/arXiv.2408.00229

© 2026 Anthony Janus. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0)